

Which One Runs Out?

Finding the limiting reactant in $\text{N}_2 + 3 \text{H}_2 \rightarrow 2 \text{NH}_3$, and working out why two reasonable-looking tests give the wrong answer.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/which-one-runs-out/

1 Before you open anything

Answer from what you already know, with the page closed. The point is to have something on paper that the demonstration can later settle.

PREDICTION

You mix **20.0 g of N_2** with **5.0 g of H_2** and let them react as far as they can: $\text{N}_2 + 3 \text{H}_2 \rightarrow 2 \text{NH}_3$. Which reactant is completely used up, and which one is still in the flask at the end?

Write one sentence saying what you compared to decide that.

2 Set up the mixture you just predicted

Open the demonstration. Two sliders set the masses you mixed; the panel headed *The three comparisons* reports that mixture three different ways.

1. Set **Nitrogen, N_2** to **20.0 g** and **Hydrogen, H_2** to **5.0 g**. (The preset button **Grams mislead** does this in one click.)
2. Copy the three comparison rows out of the table. In the last column write which reactant that row points at — the one with the *smaller* number.
3. Then record the three boxes in the **Result** panel.

Row in the table	N ₂	H ₂	Points at which reactant?
grams			
moles			
mol ÷ coef			

Limiting reactant	Theoretical yield (g NH ₃)	Left over (g, and of what)

a. Two of the three rows point at one reactant and one row points at the other. Which row disagrees with the **Limiting reactant** box, and by how much did the two masses differ — that is, how much more N₂ than H₂ did you weigh out?

b. Show the arithmetic that turns 5.0 g of H₂ into the moles value in your table, and then into the *mol ÷ coef* value. Use 2.016 g/mol.

The masses here are treated as exact, so every digit in the table is arithmetic rather than measurement. On a real problem the significant figures in the answer are set by the balance you weighed on.

3 Vary one reactant and watch the answer switch

Leave **Nitrogen, N₂** fixed at **20.0 g** for this whole section. Move only the hydrogen slider.

1. Set **Hydrogen, H₂** to each mass in the first column.
2. For each one, read the *moles* and *mol ÷ coef* entries for H₂, plus the *mol ÷ coef* entry for N₂.
3. Record the **Limiting reactant** and **Theoretical yield** boxes as well.

H ₂ (g)	mol H ₂	mol ÷ coef, H ₂	mol ÷ coef, N ₂	Limiting	Yield (g NH ₃)
2.0					
3.0					
4.0					
4.3					
4.4					
6.0					

a. One column is the same in every row. Which one, and why does it not change when you move the hydrogen slider?

b. Between which two rows does the limiting reactant switch? Write the rule you would use to decide the limiting reactant from two *mol ÷ coef* values, in your own words.

c. Adding more hydrogen raises the yield in the first few rows and then stops raising it. Give the two masses in your table between which it stops, and say what is capping the yield after that.

d. Go back to your prediction in Part 1. Was it right? If it was not, name the specific comparison you made and say what it left out. If it was right, say whether your reasoning would still have worked at H₂ = 4.0 g.

4 Using it the way a question would

Two tasks: one where the two shortcuts contradict each other, and one where you have to work backwards from the answer you want.

1. Click the preset **Knife edge** (46.3 g N₂, 10.0 g H₂). Fill in the table below.

Comparison	N ₂	H ₂	Points at
grams			
moles			
mol ÷ coef			

a. Here the grams row and the moles row point at different reactants, so at most one of them can be right. Which of the two agrees with the **Limiting reactant** box — and explain why agreeing on this mixture is not evidence that the method works.

1. Set **Nitrogen, N₂** to **50.0 g**. Before touching the hydrogen slider, calculate on paper the mass of H₂ that would consume that nitrogen exactly, with nothing left over on either side.
2. Round your answer to the nearest 0.1 g — the slider moves in steps of 0.1 g — and set **Hydrogen, H₂** to it.

b. Show the calculation, then record what the demonstration reports.

Limiting reactant: _____ Theoretical yield: _____ g Left over: _____ g of _____

c. You asked for a mixture with nothing left over, and the demonstration still names a limiting reactant and a small leftover mass. Explain why, using the fact that the slider moves in 0.1 g steps.

d. Read the **Mass check** line under the result. Write down the grams in and the grams out for this mixture, and state the law that requires them to match.

5 Solve these without the demonstration

Close the page or look away. Show your working. Then reopen it, set the sliders to the masses given, and check yourself — if a number is off, write down where. Molar masses: N_2 28.014, H_2 2.016, NH_3 17.031 g/mol.

1. 28.0 g of N_2 is mixed with 4.00 g of H_2 . Identify the limiting reactant, the theoretical yield of NH_3 in grams, and the mass of the excess reactant left over.

2. For a mixture of 60.0 g N_2 and 12.0 g H_2 , someone argues: "There are 2.14 mol of N_2 and 5.95 mol of H_2 , so nitrogen is limiting." Both mole values are correct. Explain what is wrong with the conclusion, and give the reactant that is actually limiting.

3. You have 35.0 g of N_2 and want to convert all of it, using as little excess hydrogen as possible. What mass of H_2 do you need, and what mass of NH_3 would you get? When you check, round the hydrogen up to the nearest 0.1 g so the slider can reach it.

4. A mixture contains *equal masses* of N_2 and H_2 . Without a calculator, argue which reactant must be limiting, whatever that common mass is. Your argument should use both the molar masses and the coefficients.

Every yield on this sheet is a **theoretical** yield: the ceiling if the reaction ran to completion with no side reactions. Ammonia synthesis is an equilibrium, and one pass through an industrial reactor at about 450 °C and 200 atm converts only roughly 15% of the feed, with the rest recycled. Percent yield compares a measured mass to the ceiling you calculated.

6 On your own

Nothing here is graded. The demonstration does more than this worksheet used.

- Tick *Predict first – hide the answers*, load a preset you have not used, and work the limiting reactant out from the comparison table alone before you untick it.
 - Watch the particle picture as you move the sliders. Its caption tells you how many moles one drawn molecule stands for. Find a mixture where the leftover reactant vanishes from the picture even though the mass panel still reports a leftover, and explain what the drawing is rounding.
 - The preset *All three agree* puts 80.0 g of N_2 against 4.0 g of H_2 . Starting there, find the smallest change to one slider that makes a green chip turn red.
 - Take a theoretical yield from anywhere on this sheet and work out what mass of NH_3 a single industrial pass at roughly 15% conversion would actually deliver.
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